



ENTERPRISE DATA SCIENCE BOOTCAMP

PREDICTING THE SUCCESS OF BANK TELEMARKETING CAMPAIGNS

Group 28



TELEMARKETING CAMPAIGN OPTIMIZATION:

WHO SHOULD WE CALL? AND WHEN?

THIS PROJECT IDENTIFIES BANK CUSTOMERS MOST LIKELY TO SUBSCRIBE TO LONG-TERM DEPOSIT PRODUCTS, HELPING IMPROVE CAMPAIGN EFFICIENCY AND CONVERSION RATES

AGENDA

Business Problem

Exploration and Data Analysis

Modelling Workflow

Feature Engineering

Modelling Approach

Results and Solution Output

Key Takeaways

HOW CAN BANKS TURN LOW-YIELD TELEMARKETING CAMPAIGNS INTO DATA-DRIVEN SUCCESS STORIES?



Business Problem

Telemarketing is a key channel for banks to sell long-term deposits, **but success rates are low**



Objective

Develop a predictive model to determine whether a telemarketing call will result in a **successful sale**



GOALS

- **Improve campaign efficiency** by targeting the right customers
- **Identify key factors** influencing success
- Provide **actionable recommendations** for strategy optimization



DATASET

Real-world data from a **Portuguese retail bank (2008–2013)**, including customer demographics, product details, and socio-economic indicators



STAKEHOLDERS

Marketing and sales teams aiming to enhance campaign effectiveness

OUR DATASET CAPTURES THE STORY OF CUSTOMER DECISIONS—41K CASES SHAPED BY DEMOGRAPHICS, CAMPAIGN HISTORY, AND ECONOMIC CONTEXT.



SOURCE

Portuguese retail bank marketing campaigns (2008–2013)



SIZE

41,188 records, 20 features



TARGET

y → term deposit subscription: **YES/NO**



CLASS IMBALANCE

Only ~11% positive responses

FEATURES & DATA TYPES

Customer Demographics

Integer - age

Categorical - job, marital, education, default, housing, loan

Campaign Details:

Categorical - contact, month, day_of_week

Integer - duration

History:

Integer - campaign, pdays, previous

Categorical - outcome

Economic Indicators:

Float - emp_var_rate, cons_price_idx, cons_conf_idx, euribor3m, nr_employed

Note: duration excluded from modeling ([data leakage](#))

MODELLING WORKFLOW: FROM DATA EXPLORATION TO MODEL DEPLOYMENT



EDA — Exploratory Data Analysis

Understand customer behaviour, macroeconomic influences, and campaign outcomes through visualizations and statistical analysis. These insights guide feature engineering and model selection



Baseline Modelling

A simple logistic regression is fitted with minimal preprocessing to provide a starting point for performance comparison



XGBoost Modelling

XGBoost handles nonlinearities, interactions, and mixed data types effectively. Hyperparameters and SMOTE balancing are optimized to maximize F1 score.



Deployment to Hugging Face

The selected model is wrapped inside a Gradio application and deployed to a Hugging Face Space, providing an intuitive interface for real-time predictions

1

2

3

4

5

6

7



Data Cleaning & Preprocessing

The dataset is standardized by normalizing column names, removing duplicates. This step ensures data quality and prevents leakage in later modelling



Logistic Regression Modelling

We introduce engineered features, WOE encoding, PCA, and SMOTE to improve the logistic model's ability to separate classes. This model establishes a more realistic approach for the given problem



Best Model Selection

All models are compared using their F1 scores after threshold tuning, and the best model is selected



Code and support files deployed using GIT (GIT Hub)

[20241143/EDSB_25_28](#)

KEY DATA PATTERNS GUIDED TARGETED FEATURE ENGINEERING—TRANSFORMATIONS, ENCODING, AND NEW FEATURES TO IMPROVE PREDICTIVE POWER

EDA KEY FINDINGS

ECONOMIC INDICATORS

- Strong correlation among macroeconomic variables
- Different scales require normalization

NUMERICAL FEATURES

- **High skewness and outliers** in age, campaign, previous
- **pdays behaves as binary** (mostly 999)

FEATURE IMPORTANCE

- **duration leaks target** → removed
- **month shows seasonality**
- **poutcome strongly predicts success**

HIGH CLASS-IMBALANCE PROBLEM

'No' is the dominant class (almost 90%) → **F1 score and use of Resampling Techniques (SMOTE)**



PREPROCESSING & FEATURE ENGINEERING

ECONOMIC INDICATORS

- **Applied PCA** → 2 components were kept to reduce multicollinearity

NUMERIC TRANSFORMATIONS

- pdays → **binary feature**
- month & day_of_week → **cyclical encoding**

CATEGORICAL HANDLING

- **Reduced cardinality** (job: 11→7, education: 7→5)
- Imputed missing values → **mean; median; mode**
- Encoding → **WOE encoder**

NEW FEATURES

- age × marital → interaction effect
- job × age → proxy for career stage
- campaign_log → log transformation of campaign
- contacts_ratio → ratio of contacts to previous
- economic_volatility → combined macroeconomic variability

MODELLING APPROACH: FROM A BASELINE MODEL TO A PIPELINE PROCESS

MODELLING STRATEGY

1. A **simple Baseline** model was implemented to provide a reference point for comparison

2. **Two Models were tested**, and their metrics were assessed to evaluate their performance

A **Pipeline** was built for each model:

1.Cleaning and imputation

2.Preprocessing and Data Transformation

3.Resampling (due to high class imbalance)

4.Model fitting

Model details

	BASELINE	MODEL 1	MODEL 2
Type	Logistic Regression	Logistic Regression	XGBoost
Feature Transformation	One-hot encoding	Pre-processing & Feature Engineering	Pre-processing & Feature Engineering
Reampling Techniques	None	SMOTEEN	SMOTE
Metric Criteria	F1	F1*	F1*
Hyperparameters Gridsearch: - CV: 3 - Scoring: ROC_AUC	- Max_iter= 5000 - Solver =lbfgs - Penalty = L2 - C= 1 (default)	- Preprocessor_cat_woe_regul = [0.01, 0.1, 1, 10] - Preprocessor_cat_select_k = [3, 6] - Preprocessor_num_select_k =[5, 10, 15] - Smoteen_smote_k_neighbors: [3, 5] - Smoteen_sampling_strategy = [0.5, 0,7] - Max_iter= [500, 1000, 2000] - Solver =lbfgs - Penalty = L2 - C= [0,01, 0,1, 1]	- Preprocessor_cat_woe_regul = [0.1, 10] - Preprocessor_cat_select_k = [4, 7] - Preprocessor_num_select_k =[5, 10] - Smote_k_neighbors: [3, 5] - Smote_sampling_strategy = [0.5, 0,7] - Model_max_depth = [3, 5] - Model_learnig_rate =[0.03, 0.05, 0.1] - Model_n_estimators = [200, 400] - Model_subsample = [0.8, 1] - Model_colsample_bytree = [0.7, 1] - Model_min_child_weight = [1 .5]

* after threshold tuning

RESULTS — XGBOOST ACHIEVED THE BEST OVERALL PERFORMANCE, HANDLING IMBALANCE AND CAPTURING COMPLEX PATTERNS IN THE DATA...

Features used by model

Feature	Baseline	LR Tuned	XGBoost
age	✓	✓	✗
job	✓	✓	✓
marital	✓	✗	✗
education	✓	✗	✗
default	✓	✗	✗
housing	✓	✗	✗
loan	✓	✗	✗
contact	✓	✓	✓
month	✓	✗	✗
day_of_week	✓	✗	✗
campaign	✓	✓	✗
pdays	✓	✓	✓
previous	✓	✓	✓
poutcome	✓	✓	✓
emp_var_rate	✓	✗	✗
cons_price_idx	✓	✗	✗
cons_conf_idx	✓	✓	✗
euribor3m	✓	✗	✗
nr_employed	✓	✗	✗
Engineered Features	✗	✓	✓

Evaluation metrics

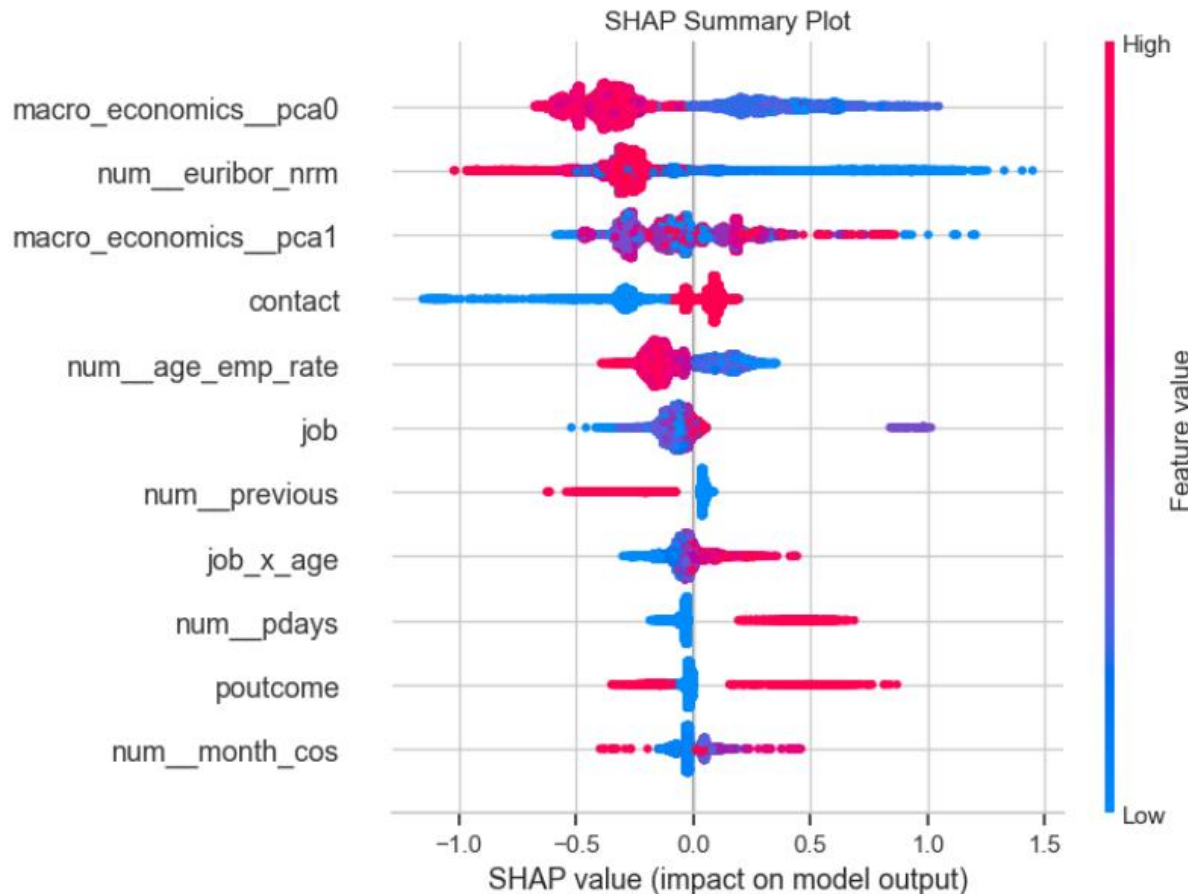
	Baseline	LR Tuned	XGBoost
ROC AUC	0.78	0.79	0.80
Accuracy	0.90	0.87	0.87
Precision	0.67	0.45	0.45
Recall	0.22	0.52	0.57
F1	0.33	0.49	0.51

BEST MODEL

- The **XGBoost model** was selected as the final production model because it achieved the highest **F1-score**, offering the best balance between Recall and Precision among all evaluated models
- For the Telemarketing Long-Term Deposit problem, the objective is two-fold:
 - 1. Maximize Recall:** Ensure that as many interested clients as possible are correctly identified, reducing missed opportunities
 - 2. Maximize Specificity:** Reduce the number of unnecessary calls to uninterested clients, improving operational efficiency and lowering campaign costs

KEY PREDICTIVE FEATURES IN THE FINAL XGBOOST MODEL

Features used by XGBOOST and their SHAP summary



The model is mostly dominated by **macroeconomic indicators**:

- **Recession periods** leads to **lower success rates**
- **Strong and stable economic periods** (lower euribor rate) tend to be **beneficial for customers**

Type of contact has influence on the campaign success rate:

- **Positive outcome** when contact is made by **Cellular**
- **Slightly negative outcome** when contact is made by **Telephone**

Age interaction with **employment variation rate** showcased relevant patterns for **targeting groups**:

- **Older individuals** even during positive employment variation are averse to commit to long term deposits
- **Younger individuals** are keen to sign long term deposits during stable economic periods

USER-FRIENDLY PREDICTIVE APP DEPLOYED FOR SALES TEAMS

- **Ready for operation** interface was deployed
- **Interactive and Practical tool** suitable for all types of technological users
- Sales Teams can use this app to evaluate if a certain Customer should be called or not

 **Bank Telemarketing Campaign Outcome Predictor:**

This application estimates the **probability** that a customer will subscribe to a term deposit based on campaign and macroeconomic features.

 **Macroeconomic Indicators**

Employment var. rate Employment variation rate - quarterly indicator	-0.1 	Consumer Price Index Consumer Price Index - monthly indicator	93.798 
			
Euribor 3 month rate Euribor 3 month rate - daily indicator	5.045 	Number of employees Number of employees - monthly indicator	5195.8 
			

 **Customer and Campaign Features**

Contact Contact communication type	telephone	Previous outcome Outcome of the previous marketing campaign	nonexistent
Job Customer's job type	technician	Age Customer's age in years	43
Days since last contact Number of days since the last contact from a previous campaign	0	Number of contacts before this campaign Number of contacts performed before this campaign for this customer	0
		Month of contact (1-12) Month when the contact was made	12

Predicted Probability

Predicted Class

Predict

PLEASE HAVE A GO!  [Bank Telemarketing Predictor](#)

KEY-TAKE AWAYS AND FUTURE WORKS

KEY - TAKE AWAYS :

- Banks can use this predictive model to choose the periods of telemarketing campaigns;
 - Avoiding calling during periods of recessions → Reducing costs on Human Resources and Unwanted Calls
 - Advantageous to perform and adjust interest rates strategies beforehand anticipating the number of customers that are willing to accept this campaign
 - Identifying target groups, mostly by age discrimination, and adapting the long-term deposit rules to their preferences

FUTURE WORKS:

- Model 2.0:
 - Group Target identification: Re-evaluation of input dataset including new attributes such as individual economical attributes
 - Testing other resampling techniques due to high level of imbalance data and/or having access to more data
- Output Integration:
 - Integrate Bank Customers DB and Macro Economic Indexes for automatic scheduling telemarketing campaign
 - Need to provide how long should telemarketing campaign run
 - Minimum number of Long- Term Deposits

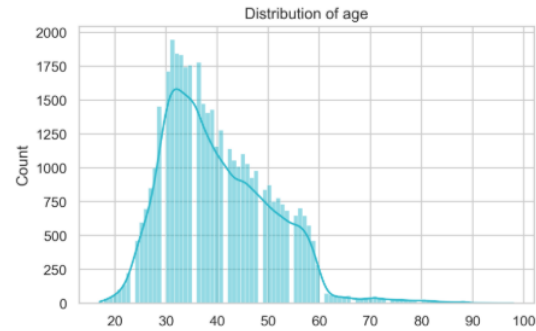
THE END

APPENDIX

NUMERICAL DISTRIBUTION

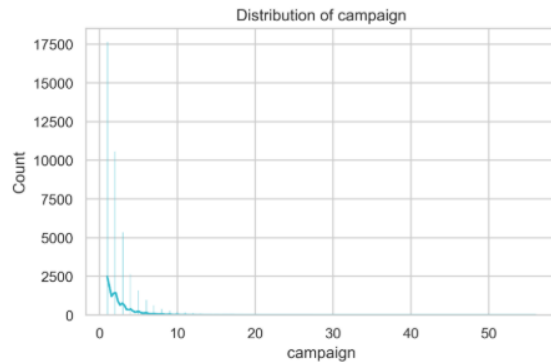
(AGE, CAMPAIGN, P DAYS)

Age



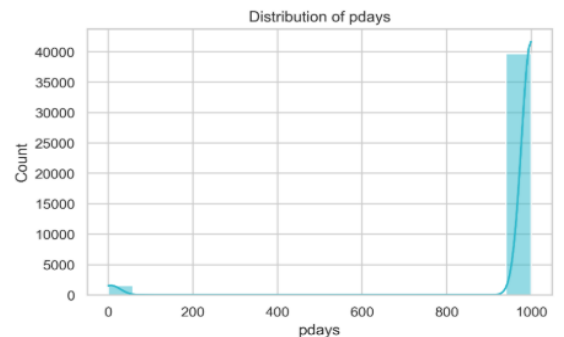
Right-skewed, concentrated roughly between 30–50 years

Campaign



Highly right-skewed; most customers are contacted only a few times, with a long tail of repeated attempts

pdays



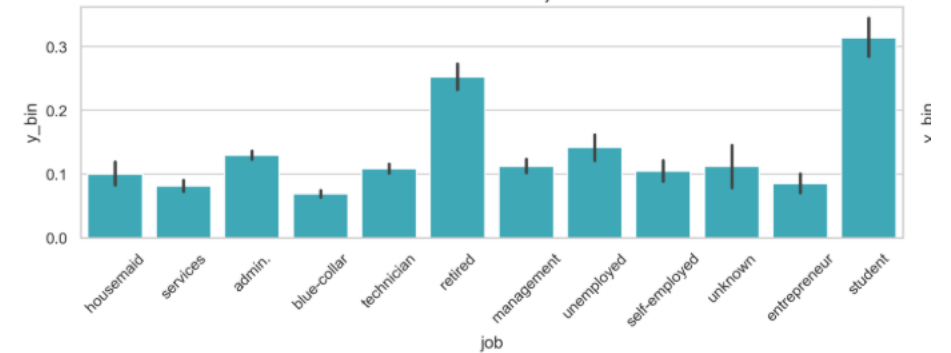
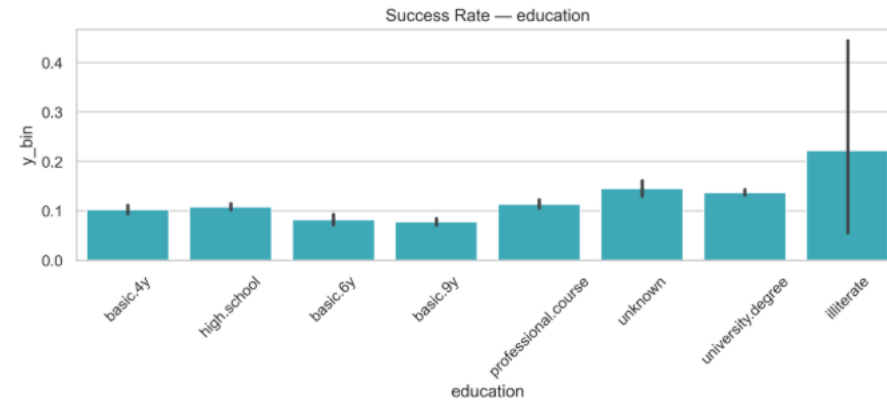
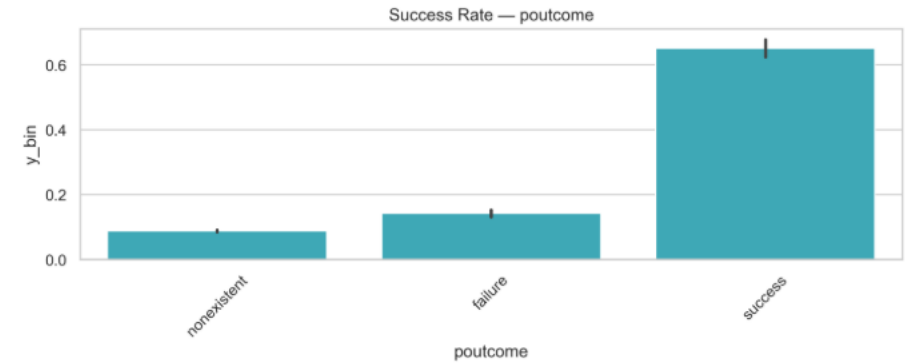
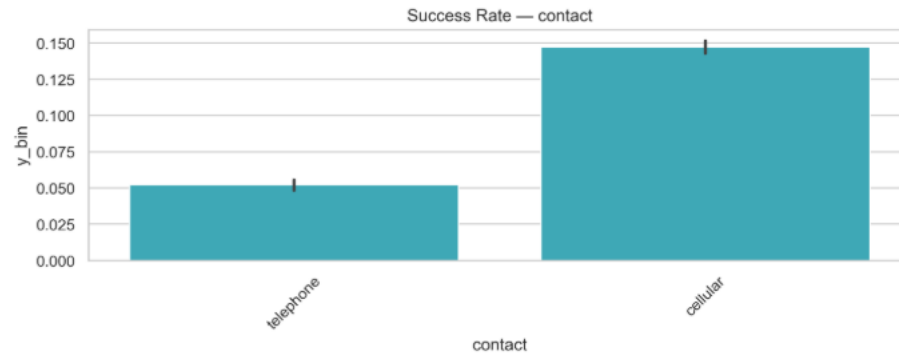
Dominated by the value 999 (no previous contact), effectively making it close to a binary indicator

SUCCESS RATE BY CATEGORICAL VARIABLE #1

(CONTACT, POUTCOME, JOB, EDUCATION)

1. Cellular and Successful outcomes in previous campaign tend to contribute to a successful subscription

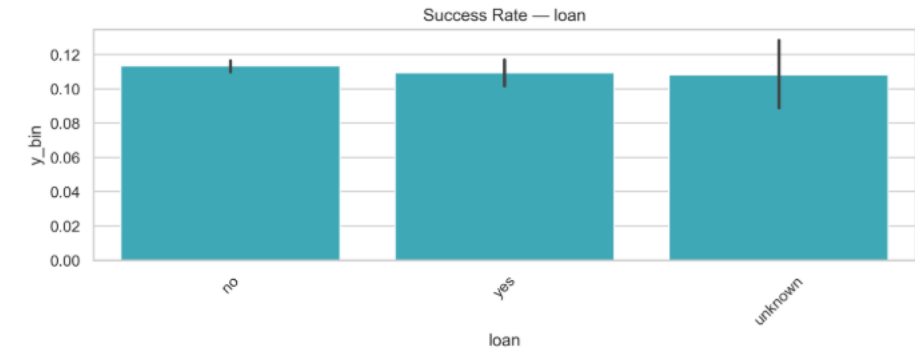
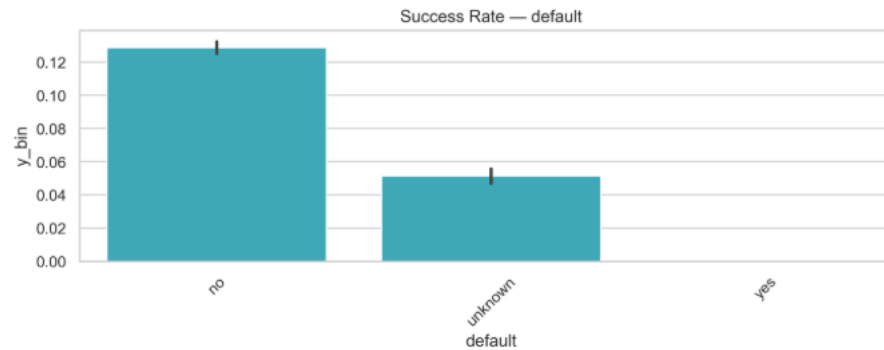
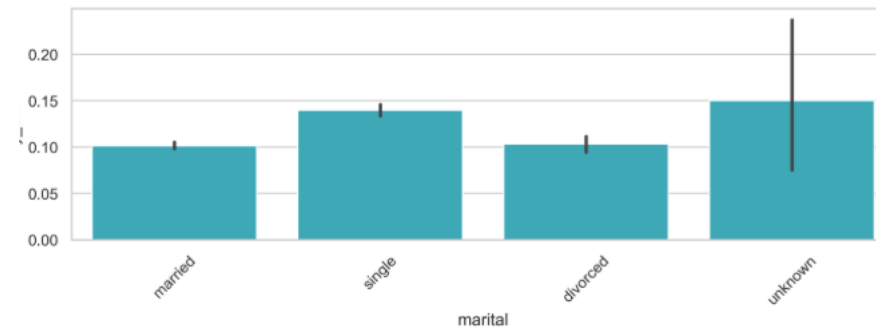
2. Student and Retire people tend to adhere more Long-Term subscriptions



SUCCESS RATE BY CATEGORICAL VARIABLE #2

(HOUSING, LOAN, MARITAL, DEFAULT)

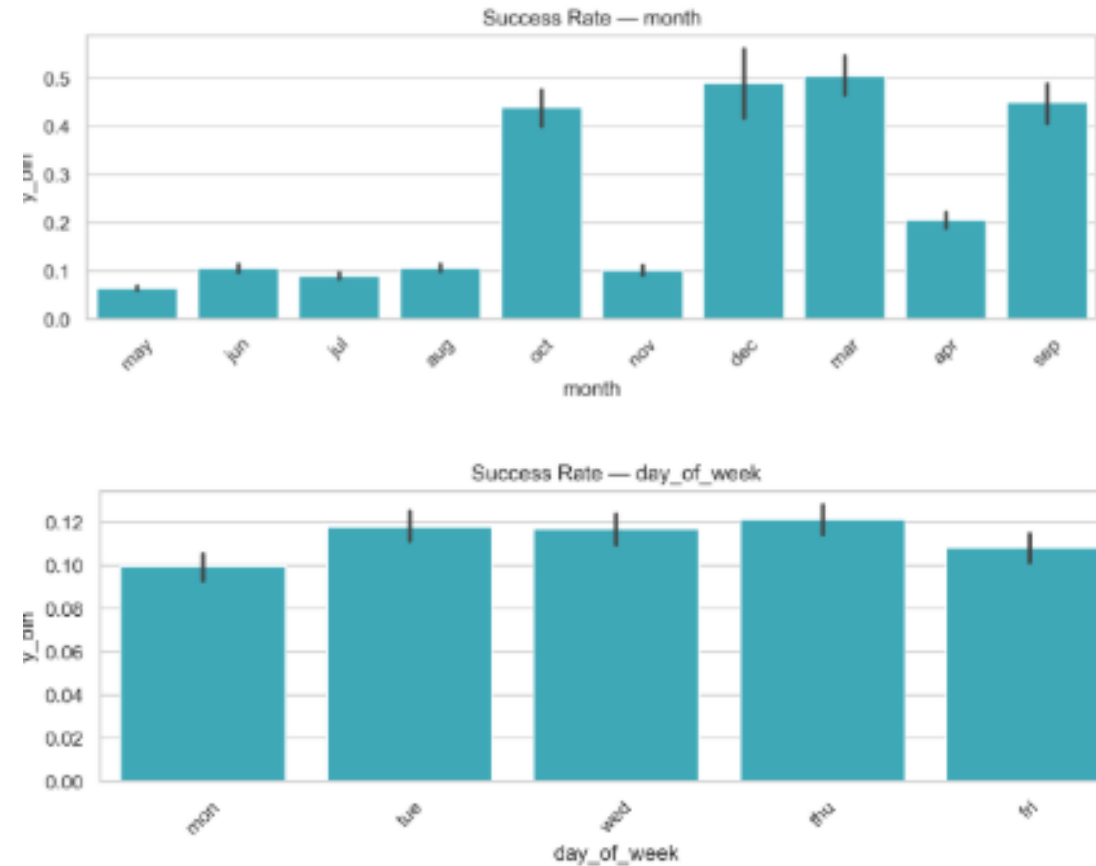
1. Distribution of Housing, Loan, marital does not show high influence on target behaviour
2. Default has no influence on target variable



SUCCESS RATE BY CATEGORICAL VARIABLE #3

(MONTH, DAY OF WEEK)

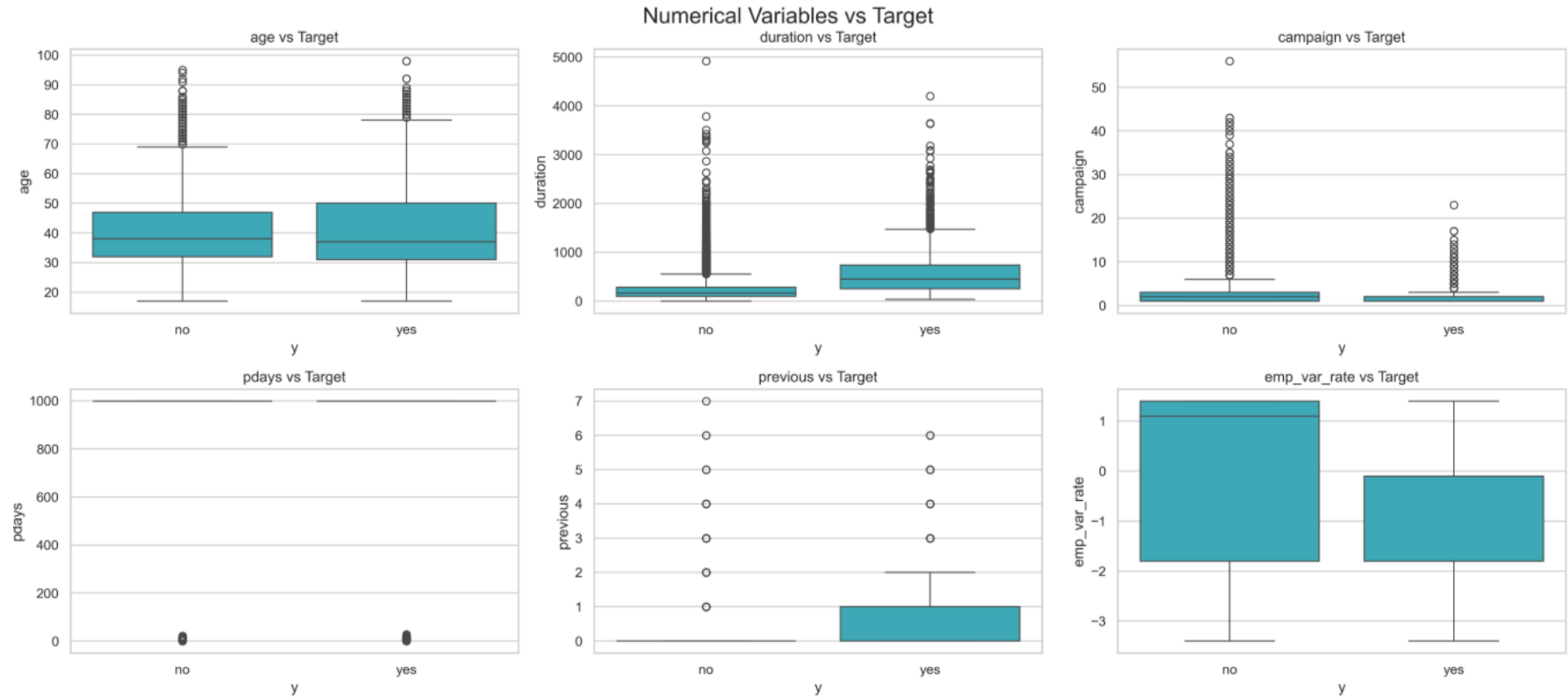
1. Uneven Distribution of success rate by Month(more prevalent) and day of the Week
2. Potentially need to assure a numerical encoding without losing cyclical characteristic



SUCCESS RATE BY NUMERICAL VARIABLES #1

(AGE, DURATION, CAMPAIGN, PDAYS, PREVIOUS, EMP_VAR_RATE)

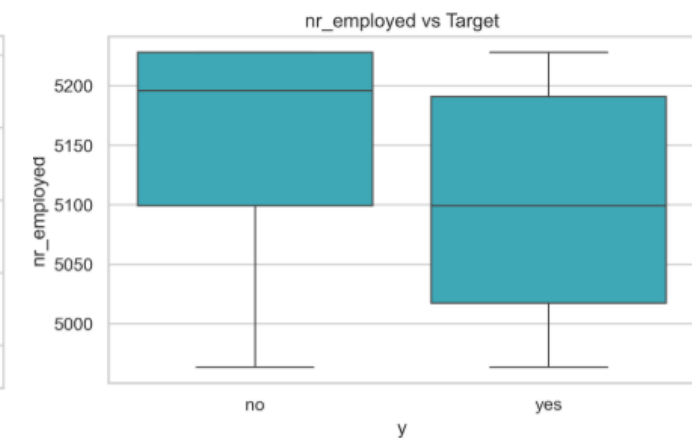
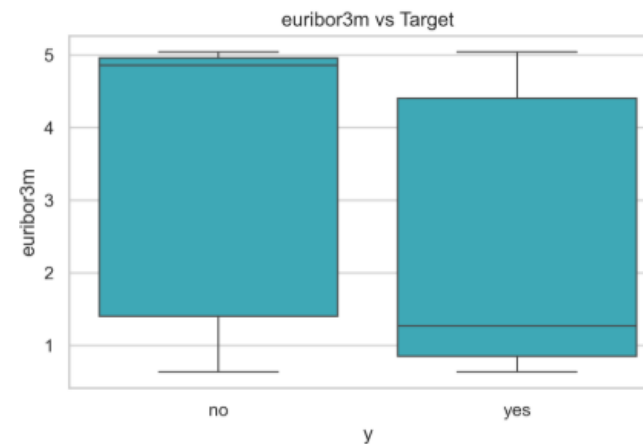
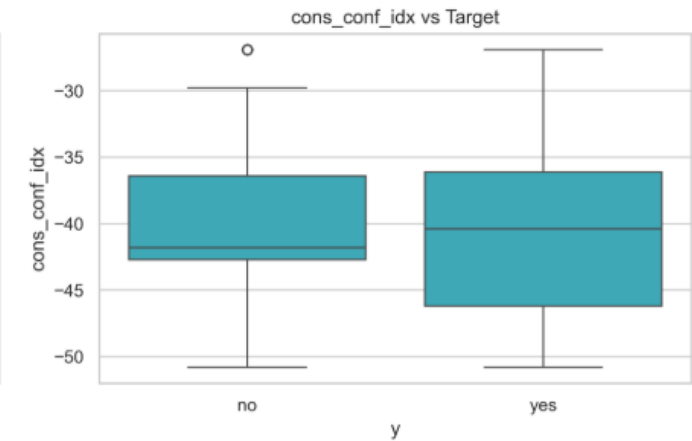
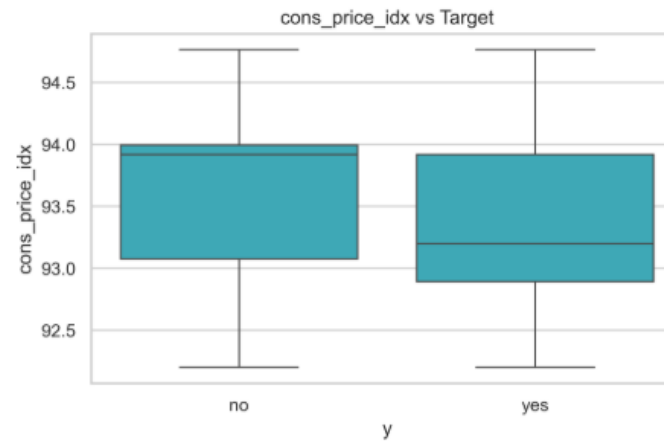
1. Higher tendency for older individuals to adhere to Long-Term;
2. Duration of call clearly influences the likelihood of campaign success:
 - Shorter calls are more likely to lead to failure;



SUCCESS RATE BY NUMERICAL VARIABLES #2

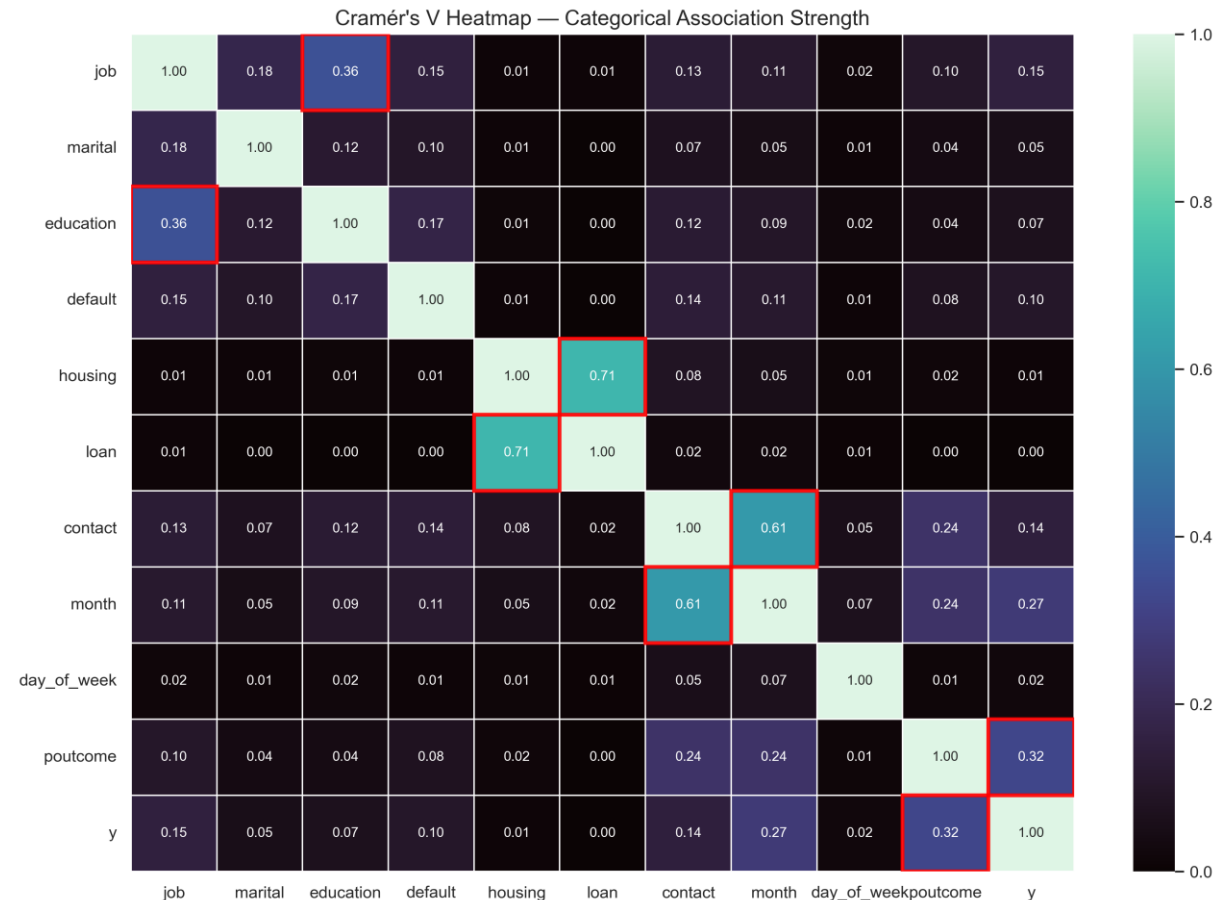
(CONS_PRICE_IDX, CONS_CONF_IDX, EURIBOR3M, NR_EMPLOYED)

1. Big variance across all macro economics variables, meaning that each factor is not discriminative by himself;
2. Euribor3m have higher discrimination power. Where positive outcomes tend to occur when euribor is low and negative outcomes when euribor rates are high;
3. Number of employed also shows to have impact on target variable;



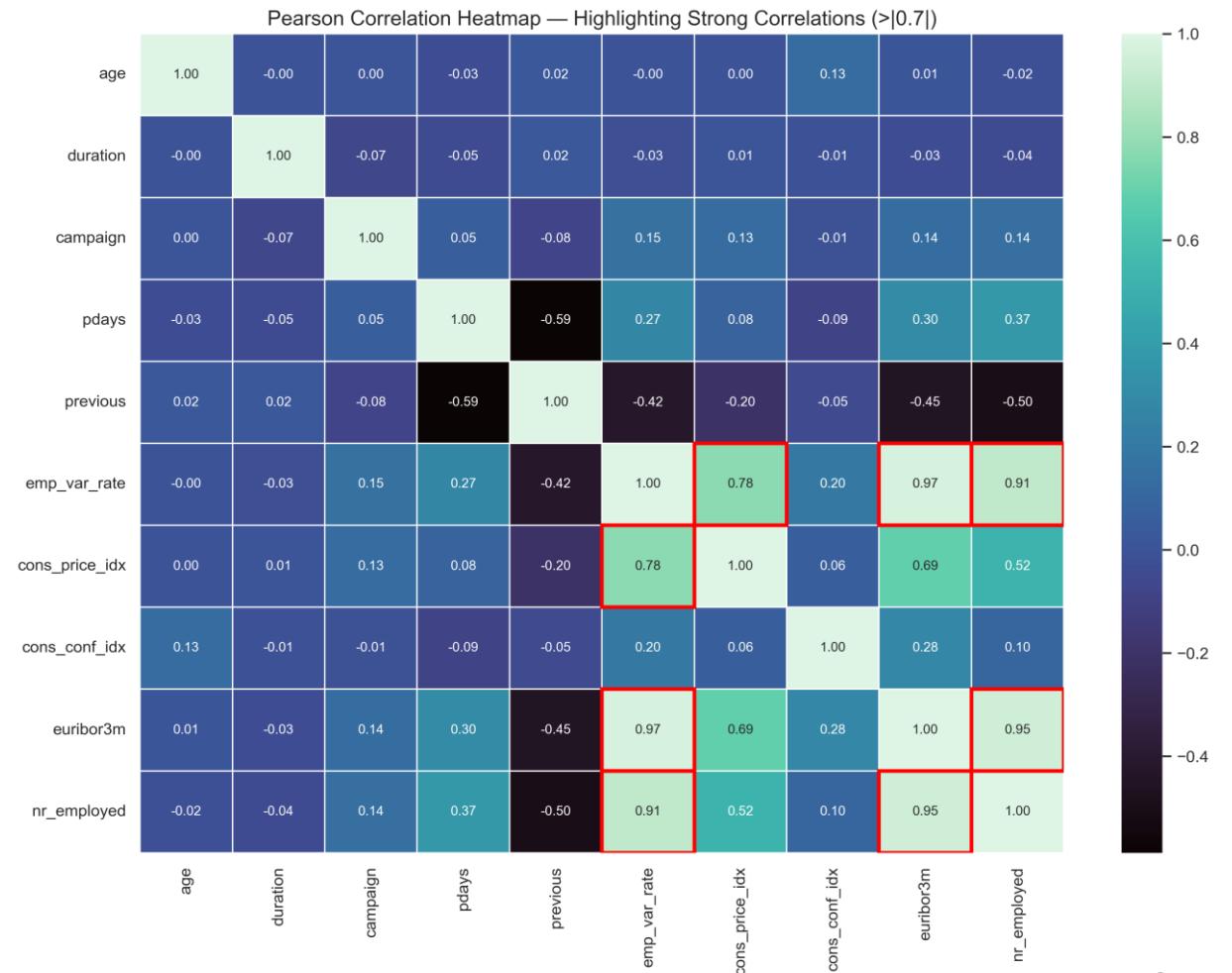
CATEGORICAL FEATURE ASSOCIATION

1. Housing vs Loan has a strong relationship (0.71) revealing that most people that contract a housing loan also adhere more to contract personal loans;
2. Contac vs Month show a high level of relationship, revealing an operational choice on when and how to call customer;
3. Poutcome and target variable have a relevant relationship demonstrating that customers with previous successful campaigns tend to adhere more to new ones;
4. Education and Job, has expected, demonstrate a relevant strength between themselves;



NUMERICAL FEATURE ASSOCIATION

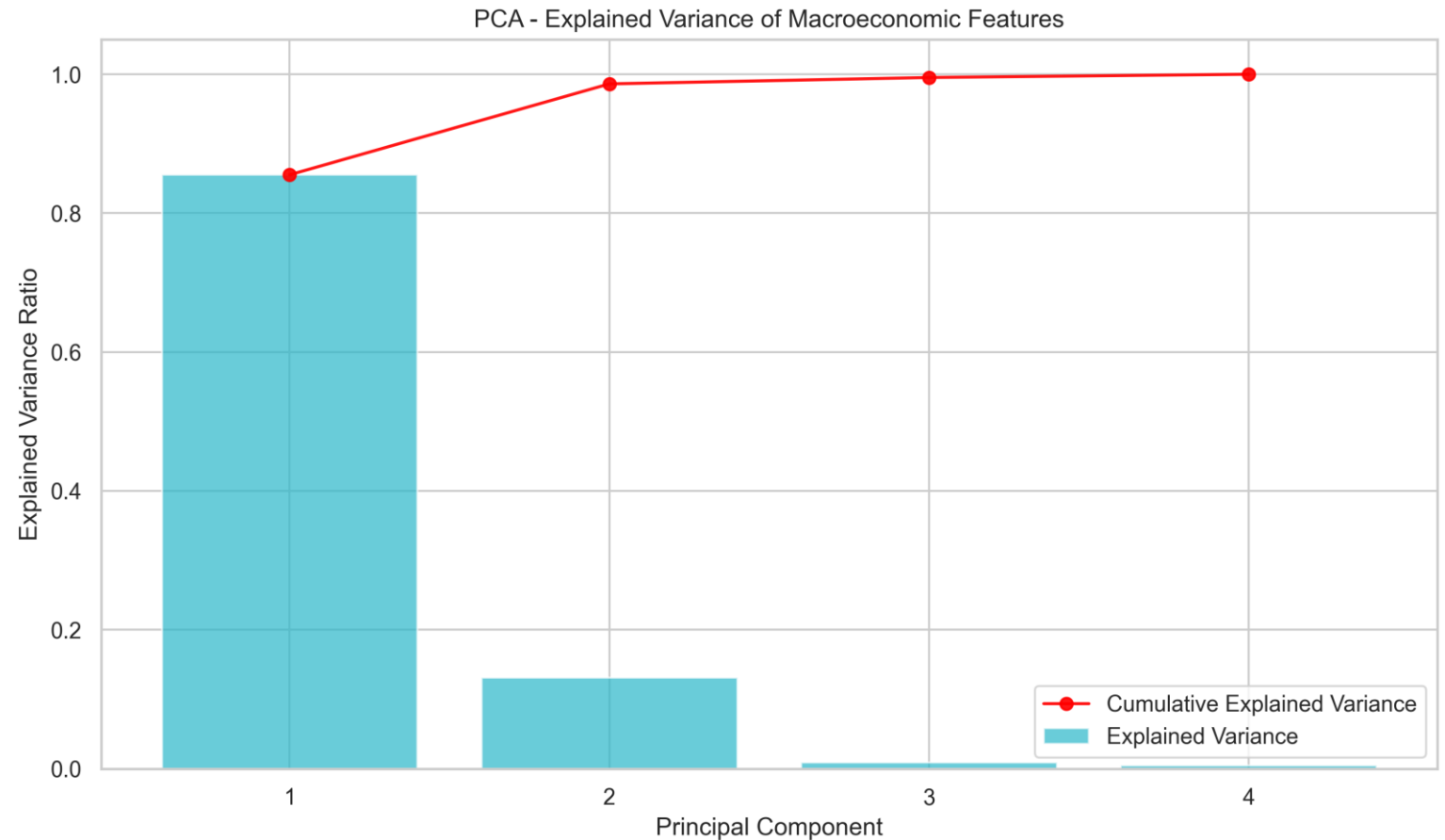
1. Most Variables are independent and mutually exclusive;
2. Strong correlation between Macro Economic Factors:
 - Use of Principal Component Analysis to reduce dimensionality could be beneficial



PRINCIPAL COMPONENT ANALYSIS

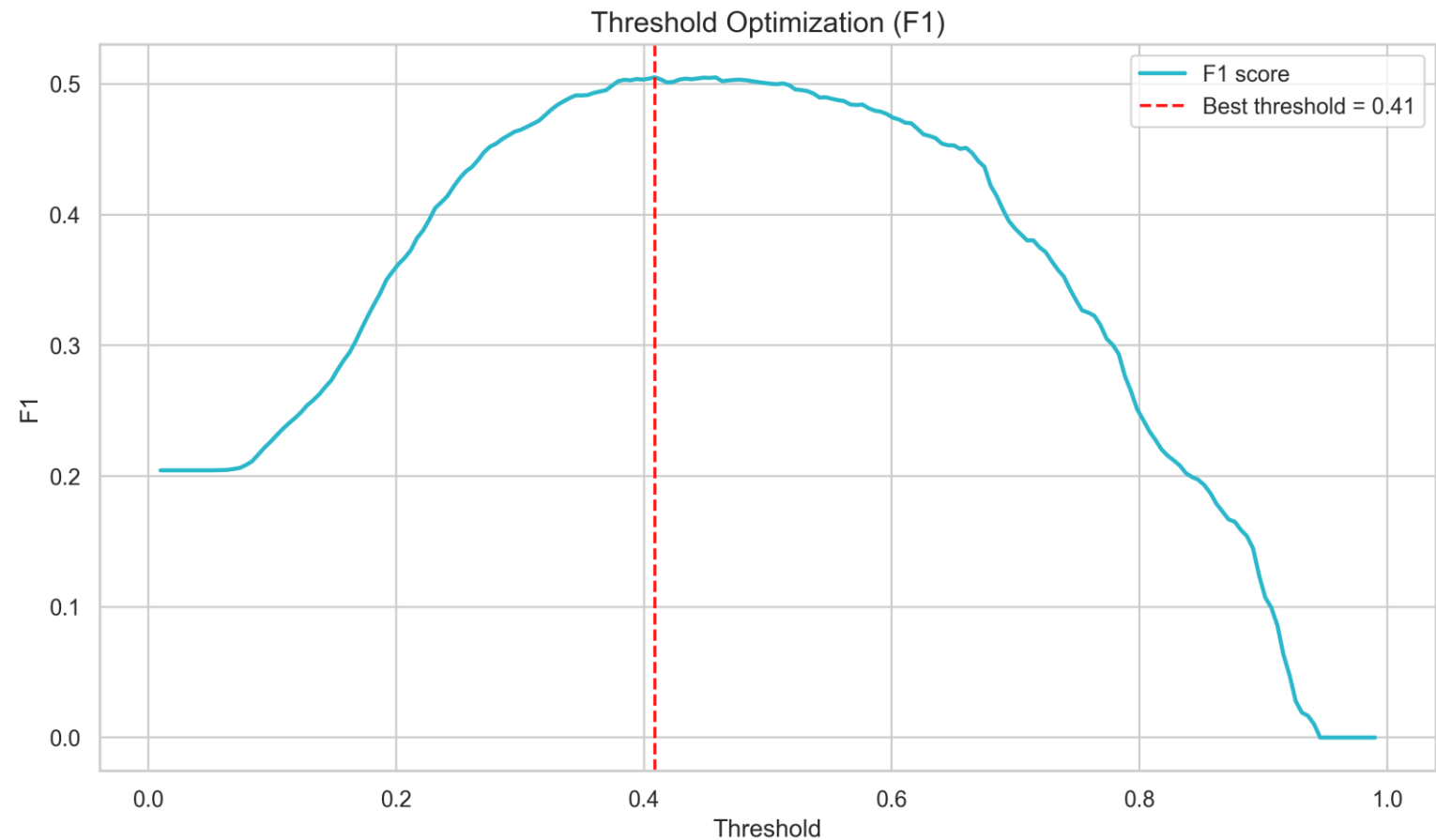
(MACROECONOMIC INDICATORS)

- Macro-economic variables are highly correlated with themselves ;
- The first 2 PCA features can explain over 90% of Explained Variance of Macroeconomic Features;
- Decision:
Use the first 2 PCA variables instead of **Euribor_3m**, **Emp_var_rate**, **cons_price_idx**, **nr_employed**



XGBOOST: THRESHOLD OPTIMIZATION TO ENHANCE F1 SCORE

- For Bank Telemarketing campaign the capability of predicting Positive Outcomes is of extreme importance (identifying when and whom to call)
- Use of optimize threshold is suitable to increase F1 score in detriment to accuracy;
- Model will predict a successful case if probability is above 0.4



XGBOOST: SHAP ANALYSIS - WHICH FEATURES INFLUENCE THE MOST...

- Macro economic variables contribute (top 3 features) contribute highly for the outcome ;
- Contact and Age_emp_Rate also have a significant impact on the prediction;
- Features like Job_Age, pdays, poutcome and month_cos are the least impactful

